Keep in Mind

This is a practice test. It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. Write each final answer in its box. Only what is inside a box is graded.

Part A — Vocabulary (8 pts)

Write the letter of the best description in the box next to each term.

1. Index **C**
2. Radicand **A**
3. Principal root **F**
4. Conjugate **H**
5. Extraneous solution **E**
6. Composition **B**
7. Inverse function **G**
8. Anchor point **D**

(A) Everything written underneath the radical symbol; when the index is even, it may not be negative.

(B) The operation that feeds one function's output into another function's input, written $(f \circ g)(x)$.

(C) The small number in the crook of the radical symbol that says which power is being undone.

(D) The point (h, k) that positions a transformed radical graph — an endpoint when the index is even, a turning point when it is odd.

(E) A candidate that satisfies the powered-up equation but fails when it is tested in the original.

(F) The non-negative value the radical symbol by itself returns; a $\pm$ must be written in by hand.

(G) The function that reverses another's inputs and outputs; its graph is the mirror image over the line $y = x$.

(H) A binomial with the same two terms but the opposite middle sign; multiplying by it turns an irrational denominator rational.

Part B — Multiple Choice (12 pts)

Write the letter of the best answer in its box.

1. Written with a rational exponent, $\sqrt[3]{x^5}$ is **A**

(A) $x^{5/3}$ (B) $x^{3/5}$ (C) x^{15} (D) $x^{-3/5}$

2. The domain of $f(x) = \sqrt{7 - 2x}$ is **C** (A) $x \geq \frac{7}{2}$ (B) $x \geq -\frac{7}{2}$ (C) $x \leq \frac{7}{2}$ (D) all real numbers

3. The graph of $y = \sqrt{3x - 12} + 5$ begins at the point **D** (A) $(12, 5)$ (B) $(4, -5)$ (C) $(-4, 5)$ (D) $(4, 5)$

4. In simplest form, $\sqrt{75} + \sqrt{12}$ is **B** (A) $\sqrt{87}$ (B) $7\sqrt{3}$ (C) $7\sqrt{6}$ (D) $10\sqrt{3}$

5. Which equation has *no real solution*? **Letter:** **C**

(A) $\sqrt[3]{x+1} = -4$ (B) $\sqrt{x-1} = 7$ (C) $\sqrt{x+6} = -2$ (D) $\sqrt{x} = 0$

6. If $f(x) = \sqrt{x}$ and $g(x) = x + 9$, then $(f \circ g)(x) =$ **D** (A) $\sqrt{x} + 9$ (B) $\sqrt{9x}$ (C) $x\sqrt{x} + 9\sqrt{x}$ (D) $\sqrt{x+9}$

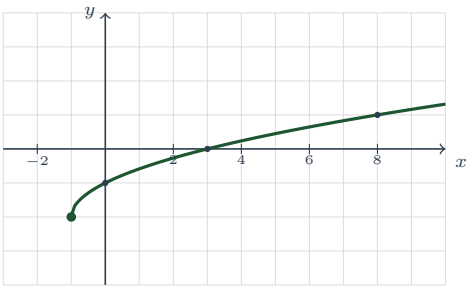
Teacher Note

Item 1: **denominator = index, numerator = power**. (B) is the signature Unit 5 error — the flipped fraction — and (C) multiplies index by power. Item 2: set the *radicand* ≥ 0: $7 - 2x \geq 0$ gives $-2x \geq -7$, and dividing by -2 **flips** the inequality. (A) is the un-flipped answer. Item 3: factor first — $\sqrt{3x - 12} = \sqrt{3}\sqrt{x - 4}$, so the graph starts at $x = 4$, not $x = 12$; (A) is the students who never factored. Item 4: $5\sqrt{3} + 2\sqrt{3}$; (A) adds radicands, (C) multiplies them, (D) multiplies the coefficients. Item 5: an isolated *even* root can never equal a negative number. Distractor (A) is the one to sort for — a student who learned "roots can't be negative" instead of "*even* roots can't" picks it, and that is a whole-unit failure, not a slip; $\sqrt[3]{x+1} = -4$ solves fine ($x = -65$). Item 6: the function next to the x runs first. (A) is the order error, (C) reads $\circ$ as multiplication.

Part C — Short Answer & Computation (40 pts)

Show your work. Assume every variable is non-negative unless told otherwise.

1. **Read the graph.** The radical function f is graphed below; each gridline is one unit and the solid dot is the endpoint.



(a) Coordinates of the endpoint: (-1,-2)

(b) Domain (interval notation): [−1, ∞); range: [−2, ∞)

(c) x-intercept: (3,0); y-intercept: (0,-1)

(d) Interval(s) where f is increasing: $[-1,\infty)$ — the whole graph

(e) Does f have an absolute maximum or an absolute minimum? Give it and say where it occurs.

absolute **minimum** -2 at the endpoint $x = -1$; no maximum

(f) End behavior: as $x \to \infty$, $f(x) \to$ ∞; as $x \to -\infty$,

nothing — the graph **stops** at $x = -1$

(g) Interval(s) where $f(x) < 0$: $[-1,3)$

2. **Radicals and rational exponents.** Rewrite or evaluate. Leave no negative exponents.

(a) Write $\sqrt[4]{x^3}$ with a rational exponent. $x^{3/4}$

(b) Write $y^{5/2}$ as a radical. $\sqrt{y^5}$, or $(\sqrt{y})^5$

(c) $27^{2/3} =$ 9 (d) $16^{-1/2} =$ 1/4

(e) Write $x^{1/2} \cdot x^{1/3}$ as a single radical. **Answer:** $\sqrt[6]{x^5}$

Add the exponents: $x^{1/2+1/3} = x^{3/6+2/6} = x^{5/6} = \sqrt[6]{x^5}$. As radicals the two factors could not be combined at all — mismatched indices — which is exactly why the rational-exponent notation exists.

3. **Simplify. Add or subtract where you can.**

(a) $\sqrt{72}$ (b) $\sqrt[3]{54}$ **(a)** $6\sqrt{2}$ **(b)** $3\sqrt[3]{2}$

(a) Largest perfect square: $\sqrt{36 \cdot 2} = 6\sqrt{2}$. (Stopping at $2\sqrt{18}$ is true but unfinished — always check the leftover.)

(b) The index sets the group size, so hunt for a perfect *cube*: $\sqrt[3]{27 \cdot 2} = 3\sqrt[3]{2}$.

(c) $\sqrt{20x^3}$ (d) $5\sqrt{12} - \sqrt{27}$ **(c)** $2x\sqrt{5x}$ **(d)** $7\sqrt{3}$

(c) $\sqrt{4x^2 \cdot 5x} = 2x\sqrt{5x}$ — divide the exponent by the index: x^3 gives one x out and one x left in.

(d) Simplify first, decide second: $5 \cdot 2\sqrt{3} - 3\sqrt{3} = 10\sqrt{3} - 3\sqrt{3} = 7\sqrt{3}$. Before simplifying these look like unlike radicals.

4. **Multiply, then rationalize.** Every answer must be in simplest form.

(a) $\sqrt{6} \cdot \sqrt{15}$ (b) $(2 + \sqrt{5})(2 - \sqrt{5})$ **(a)** $3\sqrt{10}$ **(b)** -1

(a) $\sqrt{90} = \sqrt{9 \cdot 10} = 3\sqrt{10}$. Neither factor simplifies on its own — the *product* does, so the last step is the graded one.

(b) Conjugates: $4 - 5 = -1$, a rational number. Compare $\left(2 + \sqrt{5}\right)^2 = 9 + 4\sqrt{5}$, where the radical survives.

(c) $\dfrac{6}{\sqrt{3}}$ (d) $\dfrac{4}{3 - \sqrt{5}}$ **(c)** $2\sqrt{3}$ **(d)** $3 + \sqrt{5}$

(c) Multiply by $\frac{\sqrt{3}}{\sqrt{3}} = 1$: $\dfrac{6\sqrt{3}}{3} = 2\sqrt{3}$.

(d) Multiply by the *conjugate* $\frac{3+\sqrt{5}}{3+\sqrt{5}}$: $\dfrac{4\left(3 + \sqrt{5}\right)}{9 - 5} = \dfrac{4\left(3 + \sqrt{5}\right)}{4} = 3 + \sqrt{5}$. Using $3 - \sqrt{5}$ again would only produce a new radical.

5. **Transformations.** Let $g(x) = -2\sqrt{x + 3} + 4$.

(a) List every transformation that takes $y = \sqrt{x}$ to g.

left 3, stretch $\times 2$, reflect over x-axis, up 4

(b) Anchor point: (-3,4)

(c) Domain: $[-3, \infty)$; range: $(-\infty, 4]$

(d) Is the anchor an absolute maximum or an absolute minimum? maximum ($a < 0$ turns the arm over)

(e) A different graph of the form $y = \sqrt{x - h} + k$ begins at $(5, -2)$. Write its equation. $y = \sqrt{x-5} - 2$

(f) For $p(x) = \sqrt[3]{x - 1} + 2$, give the anchor point (1,2) and the domain $(-\infty, \infty)$. In one sentence, say why that domain is not like g's.

index 3 is odd: cubing keeps the sign, so no input is forbidden

6. **Solve. Test every candidate in the original equation.**

(a) $\sqrt{2x + 3} = x$ **Answer:** $x = 3$ only

Square: $2x + 3 = x^2$, so $x^2 - 2x - 3 = 0$ and $(x - 3)(x + 1) = 0$, giving candidates $x = 3$ and $x = -1$. Check $x = 3$: $\sqrt{9} = 3$ ✓. Check $x = -1$: $\sqrt{1} = 1$, but the right side is -1, so $x = -1$ is extraneous. Solution: $x = 3$ only. (Note the tell — the non-radical side is negative at the rejected value.)

(b) $\sqrt[3]{x - 4} = -2$ **Answer:** $x = -4$

Cube both sides — legal in both directions, since every real number has exactly one real cube root: $x - 4 = -8$, so $x = -4$. Check: $\sqrt[3]{-8} = -2$ ✓. No extraneous risk with an odd index.

(c) $\sqrt{3x-2}+4=1$ **Answer:** no real solution

Isolate first: $\sqrt{3x-2}=-3$. An isolated *even* root is never negative, so no real solution — and no algebra was needed. (Squaring anyway gives $x=\frac{11}{3}$, a phantom that fails the check.)

(d) $x^{2/3}=25$ **Answer:** $x=\pm 125$

Raise both sides to the reciprocal power $\frac{3}{2}$: $x=\pm 25^{3/2}=\pm 125$. The *numerator* 2 is even — it is a square in disguise — so write the $\pm$ yourself. Both check: $(\pm 125)^{2/3}=(\pm 5)^2=25$ ✓

7. **Composition.** Let $f(x)=\sqrt{x}$ and $g(x)=x-6$.

(a) $(f\circ g)(x)=$ $\sqrt{x-6}$, domain: $x\geq 6$

(b) $(g\circ f)(x)=$ $\sqrt{x}-6$, domain: $x\geq 0$

(c) Are these the same function? Answer using *both* the rule and the domain.

No: different rules (right 6 vs down 6) and different domains ($x\geq 6$ vs $x\geq 0$)

(d) $(f\circ g)(10)=$ $\sqrt{4}=2$

(e) For $p(x)=2x+1$ and $q(x)=x^2-3$, find $(p\circ q)(-2)$. $q(-2)=1$, then $p(1)=3$

8. **Inverses.**

(a) $f(x)=4x-7$. Find $f^{-1}(x)$. $f^{-1}(x)=(x+7)/4$

(b) Verify by composition — *both* orders — that your answer really is the inverse.

$f(f^{-1}(x))=x$ and $f^{-1}(f(x))=x$: both orders return x

(c) $g(x)=\sqrt{x+2}$. Find $g^{-1}(x)$ and state its domain. $g^{-1}(x)=$ x^2-2 **domain:** $x\geq 0$

Swap and solve: $x=\sqrt{y+2}$, so $x^2=y+2$ and $y=x^2-2$. Domain: $x\geq 0$, because that is the *range* of g — read it off, never guess it.

(d) The point $(7,3)$ lies on the graph of g. What point must lie on the graph of g^{-1}? (3,7)

Teacher Note

Item 1(f) is the Unit 5 habit-breaker: "the graph stops" is a *complete* answer to the left-end question, not a missing one — do not accept "$-\infty$" or a blank. Item 1(g): the endpoint is included, so $[-1,3)$; a student writing $(-1,3)$ has forgotten that $\sqrt{0}$ is real. Item 2(e): $x^{5/6}$ alone earns most of the credit, but the item asked for a radical. Item 3: an answer that is simplified but not *fully* simplified (e.g. $2\sqrt{18}$) is half an answer. Item 4(d): watch for the a^2-b^2 sign flip in the denominator. Item 5(d) is the most-missed — a negative a makes the

anchor a **maximum**, and the range flips with it; 5(f) must name the *index*, not just assert "all reals." Item 6(a) is the graded check: "$x = 3$ and $x = -1$" as a final answer earns no credit, because the check is the step being assessed — and 6(c) should be answered *before* any squaring. Item 7(c) must cite the domain, not only the rule; "they look different" is not the argument. Item 8(b) is A2.F.2i's justification clause — both orders, or it is not a verification.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \sqrt{x-1} + 2$. **(a)** Give the anchor point, the domain, and the range, both in interval notation. **(b)** Find the x-intercept and the y-intercept, or explain why one does not exist. **(c)** List the transformations that take $y = \sqrt{x}$ to f. **(d)** Find $f^{-1}(x)$ and state its domain, saying exactly where that restriction comes from. **(e)** Justify with composition — both orders — that f and f^{-1} are inverses. **(f)** A classmate writes $y = (x-2)^2 + 1$ with *no* restriction and calls it the inverse. Explain why that is not the inverse of f.

(a) Anchor $(1, 2)$; domain $[1, \infty)$; range $[2, \infty)$. Both start at the anchor.

(b) **Neither exists.** An x-intercept would need $\sqrt{x-1} + 2 = 0$, i.e. $\sqrt{x-1} = -2$, and an even root is never negative. A y-intercept would need $f(0)$, but 0 is not in the domain — the graph does not reach the y-axis.

(c) Right 1, up 2. ($a = 1$: no stretch, no reflection.)

(d) Swap and solve: $x = \sqrt{y-1} + 2$, so $x - 2 = \sqrt{y-1}$, $(x-2)^2 = y - 1$, and $f^{-1}(x) = (x-2)^2 + 1$ for $x \geq 2$. The restriction is the **range of** f, written down: whatever f can output is exactly what f^{-1} can accept.

(e) $f\big(f^{-1}(x)\big) = \sqrt{(x-2)^2 + 1 - 1} + 2 = \sqrt{(x-2)^2} + 2 = |x-2| + 2$, and since $x \geq 2$ that is $(x-2) + 2 = x$. Going the other way, $f^{-1}(f(x)) = \big(\sqrt{x-1} + 2 - 2\big)^2 + 1 = \big(\sqrt{x-1}\big)^2 + 1 = (x-1) + 1 = x$ for $x \geq 1$. Both orders return x, so the two functions undo each other.

(f) Without the restriction it is the *whole* parabola — two arms — and it fails the horizontal line test, so it cannot be the inverse of anything: it sends both $x = 0$ and $x = 4$ to 5. It also accepts inputs f never produces; f never outputs a number below 2, so $x = 0$ has no business being fed to f^{-1}. Only the right arm, $x \geq 2$, is the inverse.

2. **Modeling.** Ignoring friction, a roller-coaster car that starts from rest and drops h feet reaches the bottom at a speed of

$$v(h) = 8\sqrt{h} \text{ feet per second.}$$

(a) State the domain that makes sense here, and connect it to the rule about what an even index will accept. **(b)** Find $v(9)$ and $v(36)$, with units. What does quadrupling the drop do to the speed, and which feature of $\sqrt{\cdot}$ is responsible? **(c)** A car reaches 56 feet per second at the bottom. Find the height of the drop algebraically, and check your answer. **(d)** Write the inverse, h as a function of v, and say what it tells an engineer that $v(h)$ does not. **(e)** Part (c) required squaring both sides. Explain why there is no danger of an extraneous solution in this problem.

(a) $h \geq 0$, i.e. $[0, \infty)$. The index is even, so the radicand may not be negative — and the context says the same thing twice over, since a drop of negative height is not a thing. Here the algebra and the situation agree; that is not always true.

(b) $v(9) = 8\sqrt{9} = 8 \cdot 3 = 24$ feet per second; $v(36) = 8\sqrt{36} = 8 \cdot 6 = 48$ feet per second. Quadrupling the drop only **doubles** the speed, because $\sqrt{4h} = 2\sqrt{h}$ — an exponent of $\frac{1}{2}$ flattens growth, which is the shape of the square root graph.

(c) $8\sqrt{h} = 56$, so $\sqrt{h} = 7$ and $h = 49$ feet. Check: $v(49) = 8\sqrt{49} = 8 \cdot 7 = 56$ ✓

(d) Solve for h: $\sqrt{h} = \frac{v}{8}$, so $h = \dfrac{v^2}{64}$ for $v \geq 0$. The original answers "how fast will this drop make the car go?"; the inverse answers the question an engineer actually asks — "how tall must I build the drop to hit a target speed?"

(e) An extraneous root only appears when squaring hides a *sign* disagreement — the step $a^2 = b^2$ forgets whether $a = b$ or $a = -b$. Here both sides are non-negative: $\sqrt{h} \geq 0$ and a speed of 56 is positive, so there is no hidden sign to lose and the single candidate must be genuine. Checking it is still good practice.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) anchor with *both* interval-notation answers; 1 pt (b) *both* non-existences with reasons — "no x-intercept" with no reason earns half; 1 pt (c) both transformations; 1 pt (d) the rule *and* $x \geq 2$ *and* the sentence that it is the range of f (no credit for the restriction if it is guessed as $x \geq 0$, the reflex answer); 1 pt (e) both compositions — one order is not a verification, and this is the standard's own "justify" clause; 1 pt (f) the horizontal line test or the two-arms argument. A student who writes $(x-2)^2 + 1$ with no restriction in (d) and then cannot explain (f) has missed the unit's closing idea — score both accordingly. D2: 1 pt (a) $h \geq 0$ tied to the even index; 1 pt (b) both values with units *and* "doubles" with $\sqrt{4h} = 2\sqrt{h}$ (half credit for "it doubles" with no reason); 1 pt (c) $h = 49$ ft; 1 pt the check; 1 pt (d) $h = \frac{v^2}{64}$; 1 pt (e) the sign argument. Do not accept "because I checked it" for (e) — the question asks why the danger was never there.